

PHYS205 Experiment 4 Lab Report: Heat Conduction

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1 Objective

The objective of this experiment is to analyze the thermal properties of different materials by observing heat conduction. This involves measuring temperature changes over time using thermistors and determining the relationship between thermal conductivity, temperature gradient, and heat flow rate.

2 Theory

In heat transfer, conduction (or heat conduction) is the transfer of thermal energy between regions of matter due to a temperature gradient. Heat spontaneously flows from a region of higher temperature to a region of lower temperature, temperature differences over time, approaching thermal equilibrium.

Fourier's Law

The law of Heat Conduction, also known as Fourier's law, states that the time rate of heat transfer through a material is proportional to the negative gradient in the temperature and to the area, at right angles to that gradient, through which the heat is flowing. We can state this law in two equivalent forms: the integral form, in which we look at the amount of energy flowing into or out of a body as a whole, and the differential form, in which we look at the flow rates or fluxes of energy locally.

Integral Form

By integrating the differential form over the material's total surface S, we arrive at the integral form of Fourier's law:

$$\frac{\partial Q}{\partial t} = -k \oint_S \vec{\nabla} T \, dA \quad (1)$$

The above differential equation, when integrated for a homogeneous material of 1-D geometry between two endpoints at constant temperature, gives the heat flow rate as:

$$\frac{\Delta Q}{\Delta t} = -kA \frac{\Delta T}{\Delta x} \quad (2)$$

where A is the cross-sectional surface area, ΔT is the temperature difference between the ends, Δx is the distance between the ends.

Differential Form

The differential form of Fourier's Law of thermal conduction shows that the local heat flux density, $\vec{q}$, is equal to the product of thermal conductivity, k, and the negative local temperature gradient, $-\nabla T$. The heat flux density is the amount of energy that flows through a unit area per unit time.

$$\vec{q} = -k \nabla T \quad (3)$$

where $\vec{q}$ is the local heat flux, Wm^{-2} , k is the material's conductivity, $\text{Wm}^{-1}\text{K}^{-1}$, ∇T is the temperature gradient, K^{-1} . For many simple applications, Fourier's law is used in its one-dimensional form. In the x-direction,

$$q_x = -k \frac{dT}{dx} \quad (4)$$

Thermistors and Steinhart-Hart Equation

A thermistor is a type of resistor whose resistance varies significantly with temperature. To accurately measure temperature, the Steinhart-Hart equation is used:

$$\frac{1}{T} = A + B \ln(R) + C(\ln(R))^3 \quad (5)$$

Where T is the temperature in Kelvin, R is resistance in Ohms, and A, B, C are the Steinhart-Hart parameters.

To measure temperature, you should measure thermistor output voltage. Voltage divider circuit is very useful to measure thermistor voltage. For this, Figure 1 set-up can use. Using voltage value, we can calculate thermistor resistance so we can find temperature value with Steinhart-Hart equation. Our sensor uses the following equation to convert resistance to temperature value.

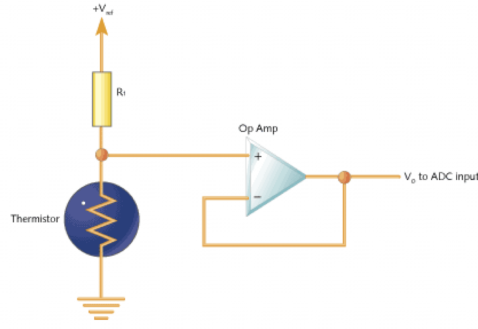


Figure 1: Thermistor value measurement method

$$T = \frac{1}{A + B \ln(R) + C(\ln(R))^2 + D \ln(R)^3} - 273.15 \quad (6)$$

For 10 KOhm Termistor (Our Sensor)

$$A = 3.354e - 3 \quad B = 2.564e - 4$$

$$C = 2.388e - 6 \quad D = 8.377e - 8$$

Table 1: Material Properties

Material	Density ρ (kg/m ³)	Specific Heat c (J/kg.K)	Conductivity k (W/m.K)
Brass (360 Alloy)	8520	385	115
Aluminum (6063-T5)	2800	830	150
Stainless Steel (304)	8000	400	14

3 Procedure

Equipment

- Heat Conduction Apparatus
- DC Power Supply
- Temperature Array Measurement Device (PIC Interface)
- 4 mm Banana Patch Cords (x2)

Experiment

1. Use two banana patch cords to connect the power supply to the Heat Conduction Apparatus. Connect the positive (+) terminal of the power supply to the red input jack of the Heat Conduction Apparatus; connect the negative (-) terminal of the power supply to the black input jack of the Heat Conduction Apparatus.
2. Connect the Temperature Array Measurement Device to a computer and launch the java based PIC software.
3. Measure the distance between the two thermistors embedded in each bar; their positions are marked by the white lines on the board.
4. Make a drawing showing the layout of the apparatus including the Peltier device, the four bars, and the thermistors. Label each thermistor with the following designations:
 - T1: Wide Brass (far)
 - T2: Wide Brass (close)
 - T3: Narrow Brass (close)
 - T4: Narrow Brass (far)
 - T5: Aluminum (far)
 - T6: Aluminum (close)
 - T7: Stainless Steel (close)
 - T8: Stainless Steel (far)

5. Place the insulators over the bars and set the switch to HEAT on the apparatus.
6. In Software, set the sampling rate of the Temperature Array to 5 Hz. Set up a graph display to show the temperature values measured by all eight thermistors.
7. Collect a test run of data to make sure that all four bars are all at room temperature. Delete the data.
8. Set the power supply to 5 V DC and simultaneously start recording. Watch the graph of the temperatures and allow them to increase for about 5 minutes. Stop recording.
9. Set the power supply voltage to zero. Remove the insulators to allow the bars to cool.

4 Data

Room temperature (T_0) = 22°C

Brass Cross-Sections:

Wide: $1.2\text{ cm} \times 0.4\text{ cm}$ Narrow: $0.8\text{ cm} \times 0.4\text{ cm}$

Other materials also have the same cross-sections as the wide brass.

Distance between sensors (Δx) = 3.0 cm

Table 2: Measured Final Temperatures ($\Delta t = 300\text{s}$)

Material	Close Sensor	T_{close} ($^\circ\text{C}$)	Far Sensor	T_{far} ($^\circ\text{C}$)	ΔT ($^\circ\text{C}$)
Aluminum	T6	36.14	T5	35.03	1.11
Wide Brass	T2	34.74	T1	32.85	1.89
Narrow Brass	T3	33.77	T4	31.25	2.52
<u>Stainless Steel</u>	<u>T7</u>	<u>32.24</u>	<u>T8</u>	<u>26.17</u>	<u>6.07</u>

Note: The bold value represents the minimum temperature difference, while the underlined value represents the maximum temperature difference.

Table 3: Heating Start Times and Propagation Delay

Material	Position	Sensor	Start Time (s)	Delay Δt (s)
Wide Brass	Close	T2	362.2	9.8
	Far	T1	372.0	
Narrow Brass	Close	T3	363.0	11.0
	Far	T4	374.0	
Aluminum	Close	T6	362.6	1.4
	Far	T5	364.0	
Stainless Steel	Close	T7	367.2	38.8
	Far	T8	406.0	

5 Data Analysis

1. Based on the T_{far} in Table 2, aluminum is the best conductor ($T_{far} = 34.31^\circ\text{C}$) because it was gotten much hotter. Stainless steel is the worst conductor ($T_{far} = 25.34^\circ\text{C}$) since it remained close to room temperature. Also, it is theoretically shown in Table 1.

2. The time delay for the wide brass bar is $\Delta t = 9.8\text{s}$ as can be seen in Table 3. We also know the distance between two thermistor as 3cm which is $\Delta x = 0.03\text{m}$.

$$v = \frac{\Delta x}{\Delta t} = \frac{0.03}{9.8} \approx 0.003 \text{ m/s} \quad (7)$$

3. The differences have already been shown in Table 2 and their conductivities was also given in Table 1.

- **Aluminum** ($k = 150$): 1.11°C .
- **Wide Brass** ($k = 115$): 1.89°C .
- **Narrow Brass** ($k = 115$): 2.52°C .
- **Stainless Steel** ($k = 14$): 6.07°C .

We can obviously see the inverse relation between k and ΔT values ($k \propto \frac{1}{\Delta T}$).

4. The ΔT initially peaks because the heat reaches the close thermistor first and heat it up while the far thermistor remains cold. This creates a large temporary difference. After the heat flow reaches the far end, the difference starts decreasing. This peak was seen first in Aluminum because it has the highest conductivity.

5. Using Formula 2 with theoretical k values in Table 1, ΔT values in Table 2, the distance between two thermistors ($\Delta x = 0.03\text{m}$) and the cross-sectional surface areas which was given at the beginning of the Data chapter:

Because we evaluate our ΔT as $T_{close} - T_{far}$, but Δx as $x_{far} - x_{close}$, $\frac{\Delta T}{\Delta x}$ seems positive but it is actually negative, so we can modify our equation with multiplied with -1 in order to use the values in tables.

$$\frac{\Delta Q}{\Delta t} = kA \frac{\Delta T}{\Delta x} \quad (8)$$

For aluminum ($k = 150$, $A = 4.8 \cdot 10^{-5}\text{m}^2$, $\Delta T = 1.11^\circ\text{C}$):

$$\frac{\Delta Q}{\Delta t} = \frac{150 \cdot (4.8 \cdot 10^{-5}) \cdot 1.11}{0.03} \approx 0.266 \text{ W}$$

For wide brass ($k = 115$, $A = 4.8 \cdot 10^{-5}\text{m}^2$, $\Delta T = 1.89^\circ\text{C}$):

$$\frac{\Delta Q}{\Delta t} = \frac{115 \cdot (4.8 \cdot 10^{-5}) \cdot 1.89}{0.03} \approx 0.348 \text{ W}$$

For narrow brass ($k = 115$, $A = 3.2 \cdot 10^{-5}\text{m}^2$, $\Delta T = 2.52^\circ\text{C}$):

$$\frac{\Delta Q}{\Delta t} = \frac{115 \cdot (3.2 \cdot 10^{-5}) \cdot 2.52}{0.03} \approx 0.309 \text{ W}$$

For stainless steel ($k = 14$, $A = 4.8 \cdot 10^{-5}\text{m}^2$, $\Delta T = 6.07^\circ\text{C}$):

$$\frac{\Delta Q}{\Delta t} = \frac{14 \cdot (4.8 \cdot 10^{-5}) \cdot 6.07}{0.03} \approx 0.136 \text{ W}$$

The highest heat flow rate is wide brass's but my intuitive expectation was that aluminum would have the highest heat flow rate but the results does not satisfies that. While it might be caused by the measurement errors and the non-ideal experimental setup, it can also be caused the given conductivity value in the experiment instruction because when I asked why my value is

out of my expectation to Gemini, it said that the conductivity constant may be mistaken and when I checked them in the Internet, I saw the conductivity value of the an 6063 aluminum as 200 W/m·K. Because of that, no correlation can be seen between ΔT and the heat flow rate as experimentally, but my expectation was an inverse correlation.

According to formula, heat flow rate is proportional to the cross-sectional area. The narrow bar has a smaller area, which causes a bottleneck while heat transfer.